Lecture 5

1. **Urea is secreted in:**
    a) PCT
    b) DCT
    c) Descending loop of Henle
    d) Ascending loop of Henle

Answer: D

2. **The prime driving force for counter current multiplier system is:**
    a) Medullary hyposmolarity
    b) Reabsorption of Na^+ in thick ascending limb
    c) Action of ADH via aquaporin channels
    d) Urea recycling

Answer: B

3. **The countercurrent exchange in the vasa recta:**
    a) removes Na^+ from the extracellular fluid.
    b) maintains high concentrations of NaCl in the extracellular fluid.
    c) raises the concentration of Na^+ in the blood leaving the kidneys.
    d) causes large quantities of Na^+ to enter the filtrate.

Answer: B

4. **The counter-current mechanism of the nephron:**
    a) is dependent on GFR.
    b) is necessary for producing concentrated urine.
    c) can be inhibited by ADH.
    d) is a function of the cortical nephrons only.
    e) both b & d are correct.

Answer: B

5. **Medullary hyperosmolarity causes:**
    a) increased Na^+ reabsorption from the DCT.
    b) increased H^+ secretion from the PCT.
    c) increased H_2O reabsorption from the collecting ducts.
    d) release of angiotensinogen.
    e) dilution of urine.

Answer: C

Lecture 6

6. **Fluid entering the loop of Henle from the proximal convoluted tubule has a higher:**
   a) Osmolarity than plasma
   b) Na^+ than plasma
   c) Glucose than plasma
   d) Ammonia than plasma

Answer: D

7. **Ammonia secretion mechanism in the kidney helps in:**
   a) Water balance
   b) Acid-base balance
   c) Temperature regulation
   d) Maintaining oncotic pressure

Answer: B

8. **The secretion of Ammonia depends upon the**:
   a) pH of tubular fluid
   b) pH of blood
   c) Renal blood flow
   d) Plasma flow

Answer: B

9. **Initial micturition is felt when the bladder gets:**
   a) Just before urination
   b) 150ml of urine
   c) 400ml of urine
   d) 600ml of urine

Answer: B

10. **Most important buffer system presents in the distal convoluted tubule:**
   a) Bicarbonate
   b) Phosphate
   c) Protein
   d) Ammonia

Answer: D

11. **Reabsorption of filtered bicarbonate or formation of new bicarbonate**:
    a) Is enhanced by carbonic anhydrase inhibitors
    b) Contributes to excretion of titratable acid
    c) Is reduced during acidosis
    d) Is accomplished by net secretion of Na^+

Answer: B

12. In cases of acidosis, the kidneys
    a) excrete urine with pH below 4
    b) increase the reabsorption of ammonia
    c) increase the reabsorption of titratable acids
    d) increase the formation of bicarbonate

answer: D

109- Which of the following occurs during acidosis:
    a) Urine pH becomes 6.5.
    b) Ammonia excretion is increased
    c) Bicarbonate reabsorption is decreased.
    d) Hydrogen secretion is decreased.

Answer: B